

Verifiable evidence for measurement reports.

Share the proof.

Keep the measurements private.

Verifiable reports across organizations.

MIDNIGHT × CLOUDFLARE

MEASURE

PROVE

VERIFY

FILMING SIMULATION

01 / 15

PRIVATE → PUBLIC

00	01	02	03	04	05	06	07
08	09	10	11	12	13	14	15
16	17	18	19	20	21	22	23

24 hourly results

Measurement values stay private

Bacchiri combines Midnight and Cloudflare to add verifiable evidence to measurement reports shared between organizations.

Share the result and its supporting evidence.

MEASUREMENT OWNER

Measurement owner

Retain sensitive records.
Explain the result.

Public evidence

VERIFIER

Verifier

Date, conditions, result.
Check the supporting evidence.

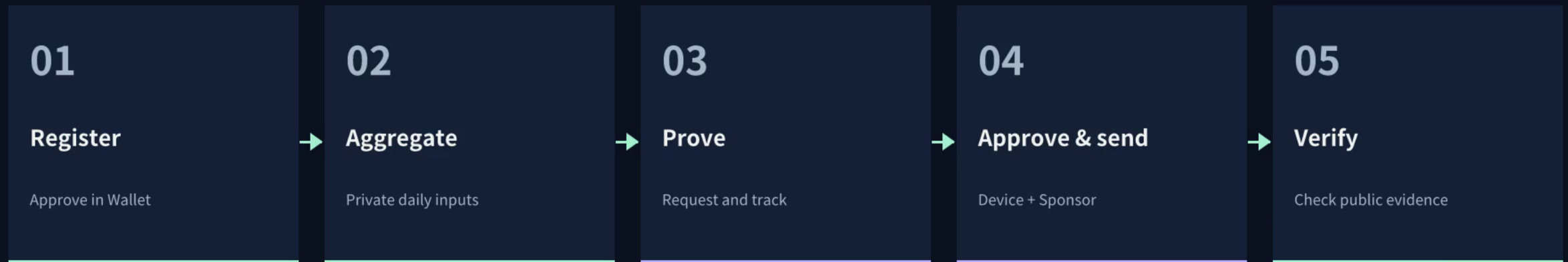
Existing devices + existing cloud services

From construction-site reporting to a partner pilot.

Our first business focus is measurement reporting on construction sites, followed by a partner pilot.

From registration to independent verification.

Connect your Wallet. Review and approve.



Five use cases. One connected workflow.

These five use cases form one connected workflow.

Bind the Device to its conditions first.

01

Link Device to its Policy.

WHAT THIS STEP ESTABLISHES

Authenticate with Wallet

Approve registration

Assign the Device Policy

Check registration and assignment

FILMING SIMULATION

04 / 15

RECORDED DEMO

AUTHENTICATE / REVIEW / REGISTER

2. Midnight registration

Policy

Cold storage 2-8 °C / 2-8 °C

2-8 °C

DEMO-POLICY-TX

✓ Register Device and assign policy

✓ Device registration and Policy assignment complete (simulation)

Registration Job	demo-registration-8
Status	completed
Device registration TX	DEMO-DEVICE-TX
Threshold assignment TX	DEMO-ASSIGNMENT-TX

Contract: SIMULATED-CONTRACT-NO-LEDGER

One day. Twenty-four hourly slots.

02

Prepare proof.
Preserve privacy.

PRIVATE INPUTS / 24 HOURS

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Share only the required results.

Separate values from public results

RECORDED DEMOAGGREGATE / CHECK INPUTS / SET BOUNDARIES

4. ZK proof and Midnight record

▶ Submit ZKP generation and TX processing

Proof Job

—

Status

—

Processing status: 2026-09-10: Sensor readings used: 1440, Outliers: 0, Status: WITHIN THRESHOLD

TX: —

TRANSACTION FEE SPONSORED

The service Sponsor Wallet adds the DUS

Daily sponsored submissions (JST)

100 remaining / 100 (0 used)

Resets: 13/09/2026, 19:00:43

Daily sensor history

Period	Sensor readings used	Outliers	Status
2026-09-10 View this day	1440	0	WITHIN THRESHOLD Collecting readings Private proof input available

2026-09-10 — Hourly min / max / average

Period	Count	Minimum	Maximum	Average
10/09/2026, 00:00:00 10/09/2026, 01:00:00	60	4.6 °C	5.4 °C	
10/09/2026, 01:00:00 10/09/2026, 02:00:00	60	4.73 °C	5.53 °C	5.1

Raw measurements and actual hourly values stay separate from the public output.

From request to proof and transaction preparation.

03

Request.
Track. Approve.

WHAT THIS STEP ESTABLISHES

- Proof Job ID
- Proof and transaction preparation

Check transaction progress

RECORDED DEMOREQUEST / TRACK THE JOB / APPROVE

Processing status: Simulated confirmation complete: DEMO-TX-demo-proof-11
TX: DEMO-TX-demo-proof-11

TRANSACTION FEE SPONSORED
The service Sponsor Wallet adds the DUST fee and submits the approved Device transaction.

Daily sponsored submissions (JST)
99 remaining / 100 (1 used)
Resets: 13/09/2026, 19:01:05

	Status	Action
0	WITHIN THRESHOLD Simulated confirmation Private proof input available	Verify daily ZKP

Use the Job ID to follow acceptance, proof generation, and transaction preparation.

Your Wallet approves. The Sponsor pays the fee.

04

Separate approval from fee payment.

FROM APPROVAL TO PUBLIC RECORD

Device Wallet

Review and approve



Sponsor Wallet

Add DUST and submit



Sponsor adds the DUST fee

FILMING SIMULATION

07 / 15

RECORDED DEMO

APPROVE / SPONSOR THE FEE / TRACE THE TX

Processing status: Simulated confirmation complete: DEMO-TX-demo-proof-11
TX: DEMO-TX-demo-proof-11

TRANSACTION FEE SPONSORED
The service Sponsor Wallet adds the DUST fee and submits the approved Device transaction.

Daily sponsored submissions (JST)
99 remaining / 100 (1 used)
Resets: 13/09/2026, 19:01:05

	Status	Action
0	WITHIN THRESHOLD Simulated confirmation Private proof input available	Verify daily ZKP

Verify daily ZKP

The service's Sponsor Wallet adds the DUST fee to that approved transaction and submits it.

Verify the result using public evidence.

05

Check the proof.
Keep values private.

WHAT THIS STEP ESTABLISHES

- Shared transaction identifier
- Match date, Policy and record
- Hourly results

Public output: hourly results

FILMING SIMULATION

08 / 15

[illegible]

The verifier can confirm the result and its supporting evidence without receiving the original measurement values.

DISTINCT PUBLIC OUTCOMES

Condition not met

Information unavailable

Invalid input rejected

FILMING SIMULATION

09 / 15

RECORDED DEMO

OUTSIDE / MISSING / REJECTED

NO DATA
NO DATA
NO DATA
NO DATA
NO DATA
NO DATA

match the public

IN THRESHOLD

 Transaction found on Midnight
de0000000000...000000000019

 Public verification complete
4 / 4 checks complete

Verifiers can distinguish a failed condition from a lack of information.

API

WHAT THIS STEP ESTABLISHES

- Register source and Policy
- Authorized service Wallet
- Share verifiable results

FILMING SIMULATION

10 / 15

RECORDED DEMO

CONNECT / TRACK / SHARE

SIMULATED • CONFIRMED

✓ ZK proof + System Wallet

✓ Midnight confirmation

The proof discloses hourly outcomes and the public Policy, not individual readings.

Maximum	Average	Threshold result
5.4	5.00	● WITHIN-THRESHOLD
5.53	5.13	● WITHIN-THRESHOLD
5.65	5.25	● WITHIN-THRESHOLD

The service Wallet performs authorized operations, adding public verification to an existing measurement workflow.

From service health to processing events.

OPS

Support reliable
daily operations.

WHAT THIS STEP ESTABLISHES

Service and Wallet health

Daily metrics and events

Investigate jobs needing attention

Inspect the action and outcome

FILMING SIMULATION

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RECORDED DEMO

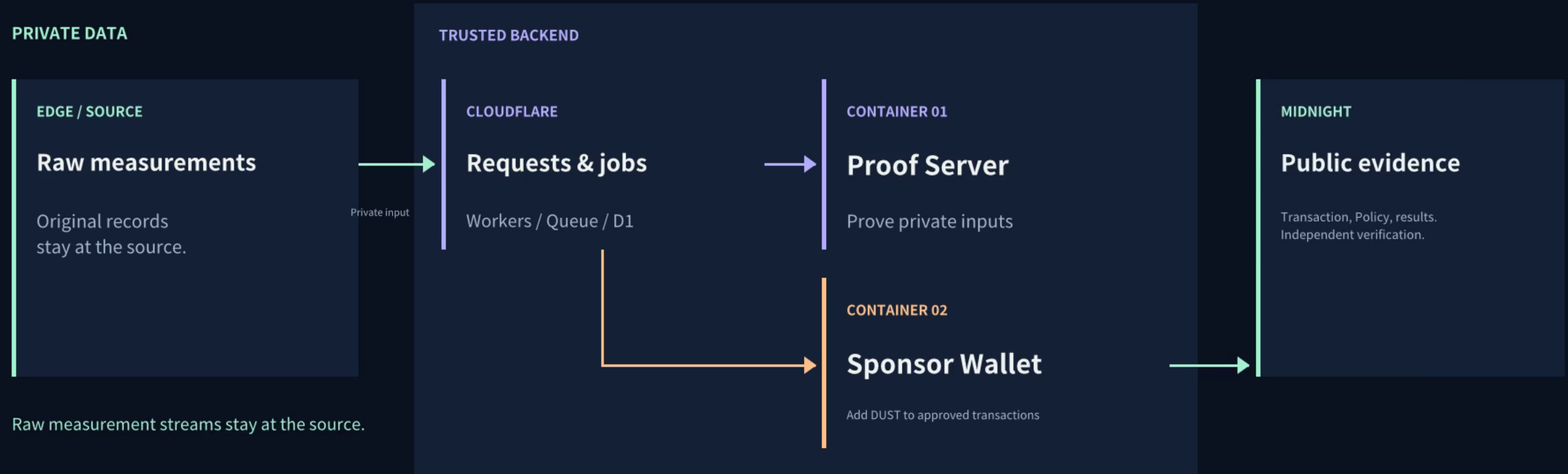
HEALTH / METRICS / EVENTS

Event details

Timestamp	2026/09/12 19:03:14
Category	workflow
Severity	info
Actor type	demo
Actor identifier	LOCAL SIMULATION
Action	simulated_run_retried
HTTP	—
Outcome	success
Request ID	—
Client Operation ID	—
Project	demo-project-1
Device	—
Resource	—
Correlation ID	—

Operational information stays separate from secrets, while giving the team what it needs to keep the service running.

Separate request handling, proof generation and Wallet.



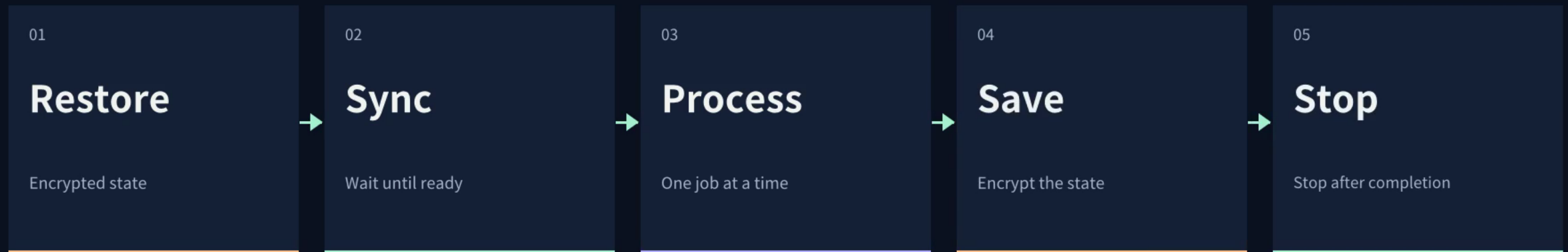
Raw measurement streams stay at the source.

Private proof inputs are processed by the trusted backend.

Private proof inputs are processed by the trusted backend.

Manage the Wallet from restore to shutdown.

Start when there is work to do.



02:00

Example daily processing start

A start time is not a guaranteed confirmation time.

It does not guarantee when a transaction will be confirmed.

Be clear about proof and trust.

PROVEN RELATION

What the proof establishes

Private hourly inputs
× registered conditions

+

SEPARATE ASSURANCE

Separate assurances

Physical sensor accuracy.
Complete measurement and correct aggregation.

CURRENT TRUST BOUNDARY

The current design trusts the backend and proof service.

Next: a partner pilot.

Evaluate value in a real workflow.

EVALUATE 01

Reliability

Can daily verification be sustained?

EVALUATE 02

Support effort

What does setup and operation require?

EVALUATE 03

Customer value

Does it improve decisions and work?

Agree on the use case, evaluation period and success metrics.

Assess continued use and commercial delivery.

We will agree on the temperature measurement use case, evaluation period, and success metrics, then assess continued use and the conditions for commercial delivery.

BACCHIRI!

VERIFIABLE MEASUREMENT

**Trust in data.
Proof for everyone.**

Thank you.

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Backlist available Midnight and Cloudflare to add proof to

FULL DEMO — Play the embedded movie in PowerPoint or open the MP4.